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Spectroscopic Signature of r-Process Nucleosynthesis in a Gamma-Ray Burst: Possible Resonant Deuterium Absorption in GRB 210704A

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ABSTRACT

Catastrophic stellar endpoint events, such as binary neutron star mergers and collapsars, are proposed to be sites of rapid neutron-capture process (r-process) nucleosynthesis; for neutron star mergers in particular, this is typically inferred from late-time kilonova emissions. We report the detection of a statistically significant spectral gap between sub-GeV and GeV photons in the prompt emission of GRB 210704A. By constraining the bulk Lorentz factor, we map this gap to an energy of $\sim 2-7$ MeV in the comoving frame. We show that this feature is naturally explained by resonant photodissociation of deuterium (^{2}H), whereas the Giant Dipole Resonance of heavier nuclei occurs at much higher energies ($\sim$ 20 MeV). Furthermore, the extremely short dynamical timescale of prompt emission favors the survival of deuterium, as the formation of heavier r-process elements is bottlenecked by slower β-decays. Considering both possible origins of this burst, a collapsar ($z \simeq 2.34$) and a compact-object merger ($z \simeq 0.27$), we demonstrate the plausibility of this unusual observation in GRB 210704A: the baryon loading in the outflow of this burst could provide a sufficient optical depth for photodissociation, while the optical depth of Compton scattering for GeV photons drops significantly in the Klein-Nishina regime for the electrons accelerated by internal shocks, so that GeV photons are emitted while those in the resonant region are absorbed by deuterium. This observation provides the first spectroscopic evidence of light r-process element formation within a relativistic jet, linking the immediate aftermath of the relativistic outflow generated in a catastrophic stellar endpoint to nucleosynthesis.

Keywords: Particle astrophysics (96); High energy astrophysics (739); Particle physics (2088); Gamma-ray bursts (629)

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1. INTRODUCTION

Core-collapse supernovae (CC SNe) and binary neutron star (BNS) mergers both create neutron-rich environments; they are the two leading candidates for producing most of the rapid neutron-capture process (r-process) elements in the universe (e.g. Lattimer & Schramm 1974; Eichler et al. 1989; Arnould et al. 2007; Argast et al. 2004; Matteucci et al. 2014; Wehmeyer et al. 2015; Radice et al. 2018; Metzger 2019; Wang et al. 2020; Levan et al. 2024). Although the gravitational wave event GW170817 and its associated kilonova AT2017gfo confirmed that such mergers synthesize heavy elements (Abbott & et al. 2017; Metzger 2019), observational evidence has largely been restricted to the thermal emission of radioactive decay on timescales of days. High-enthalpy winds and fireballs associated with gamma-ray bursts (GRBs) have been proposed to be a source of light-element synthesis (e.g. Fowler et al. 1967; Pruet et al. 2001; Otsuki et al. 2000; Thompson et al. 2001; Pruet et al. 2002). This suggests that nuclear processes are highly likely to occur in GRB formation scenarios. However, investigating nucleosynthesis during the *prompt* phase (minutes after the burst) remains a significant challenge, primarily due to the intense radiation masking nuclear spectral lines.

The stellar progenitor of GRB 210704A remains under debate (Malacaria et al. 2021; Becerra et al. 2023; Pieterse et al. 2026): in the recent study of Pieterse et al. (2026), a broad dip at $\approx$ 4050 Å in the afterglow spectrum is interpreted as Ly-α absorption, which implies a collapsar nature with a redshift of $z = 2.34$, whereas Becerra et al. (2023) note that this feature occurs in a rather noisy part of the spectrum, point out the absence of the corresponding C IV and Si IV absorption, and propose a merger origin with $z < 0.4$. Meanwhile, the burst's prompt high-energy emission—central to the present study—has not been rigorously characterized in terms of its phenomenological features. In this *Letter*, we focus exclusively on data-driven observational findings, i.e. the statistically significant gap in the high-energy spectrum; we demonstrate that this anomaly corresponds to the photodissociation threshold of deuterium (^{2}H) in the jet's comoving frame in both hypotheses.

The paper is organized as follows: Section 2 describes key properties of prompt emission of GRB 210704A and spectral anomaly in high-energy prompt emission; the redshift and Lorentz Factor are estimated. Section 3 presents a quantitative statistical analysis to confirm the existence of gaps between the sub-GeV and GeV photons from the time-integrated (TI) and time-resolved (TR) spectra. Section 4 presents candidate physical interpretations derived from observational constraints, of which the photodissociation of deuterium is most natural and reasonable. In Section 5, the plausibility for the observation of photodissociation of deuterium in GRB 210704A is discussed; their implications for future observational studies of prompt high-energy emission of GRB are discussed.

2. THE OBSERVATION OF GRB 210704A

The anomaly in high-energy prompt emissions.— The prompt high-energy emission ($>$ 100 MeV) peaks at $T_0 + 0.96$ s, with the highest-energy photon (denoted by $\gamma_{16.1\mathrm{GeV}}$) as shown in Figure 1 (a). The light curve of the high-energy emission traces that of the MeV prompt emission almost simultaneously, or with a very small time delay, which implies that it may mainly have an internal origin (Maxham et al. 2011; Pe'Er et al. 2012). The LAT data of the TRANSIENT020 class (or evclass=16) are used in the analysis to maximize the statistics for the spectral analysis of the prompt emission of GRB. [1]

From the energy distribution of the first 2 s as shown in Figure 1 (a), the upper energy envelope of sub-GeV photons begins at 0.55 GeV (denoted by $\gamma^{\mathrm{envelope}}_{\mathrm{sub-GeV}}$) at $T_0 + 0.7$ s, drops to 0.25 GeV at $T_0 + 1.2$ s, and then increases to 0.6 GeV after $T_0{+}2$ s. It seems that the lower energy envelope of GeV photons (denoted by $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$) evolves with a similar trend, while the intersection region of all gaps is observed between 0.62 GeV and 1.21 GeV.

Light curves and spectrum of prompt emission.— GRB 210704A has an intermediate duration, and most of the radiated energy of MeV photons is emitted in the first 2 s. Its prompt high-energy emission ($>$ 100 MeV) arrived almost simultaneously with the MeV emission, as shown in Figure 1 (a).

The Markov Chain Monte Carlo (MCMC) fitting is performed to find the parameters with maximum likelihood for the spectrum. The sampling tool is EMCEE (Foreman-Mackey et al. 2013). From the modeling results with empirical functions (e.g., the BAND function and exponential cutoff power-law, i.e. CPL), the low-energy photon index (α) of the spectra varies from 0 to -0.64 within the first 2 s, as shown in Figure 1(a); after $T_0 + 2$ s, α falls below -1.0. In modeling of the spectrum of the first 0.5 s, $\alpha \simeq 0$; by the Bayesian information criterion method (BIC, Wei et al. 2016), CPL is better than the BAND function in modeling, which means the high-energy photon index $\beta \to -\infty$. This implies a typical photospheric emission in the beginning of the MeV prompt emission.

A combined model of multicolor blackbody (mBB, e.g. see details in Pe'er 2008; Ryde et al. 2010; Hou et al. 2018) plus a BAND function for the broad energy band is used to describe the spectrum of prompt emission from

[1] see the details on the web page in Fermi LAT caveats.

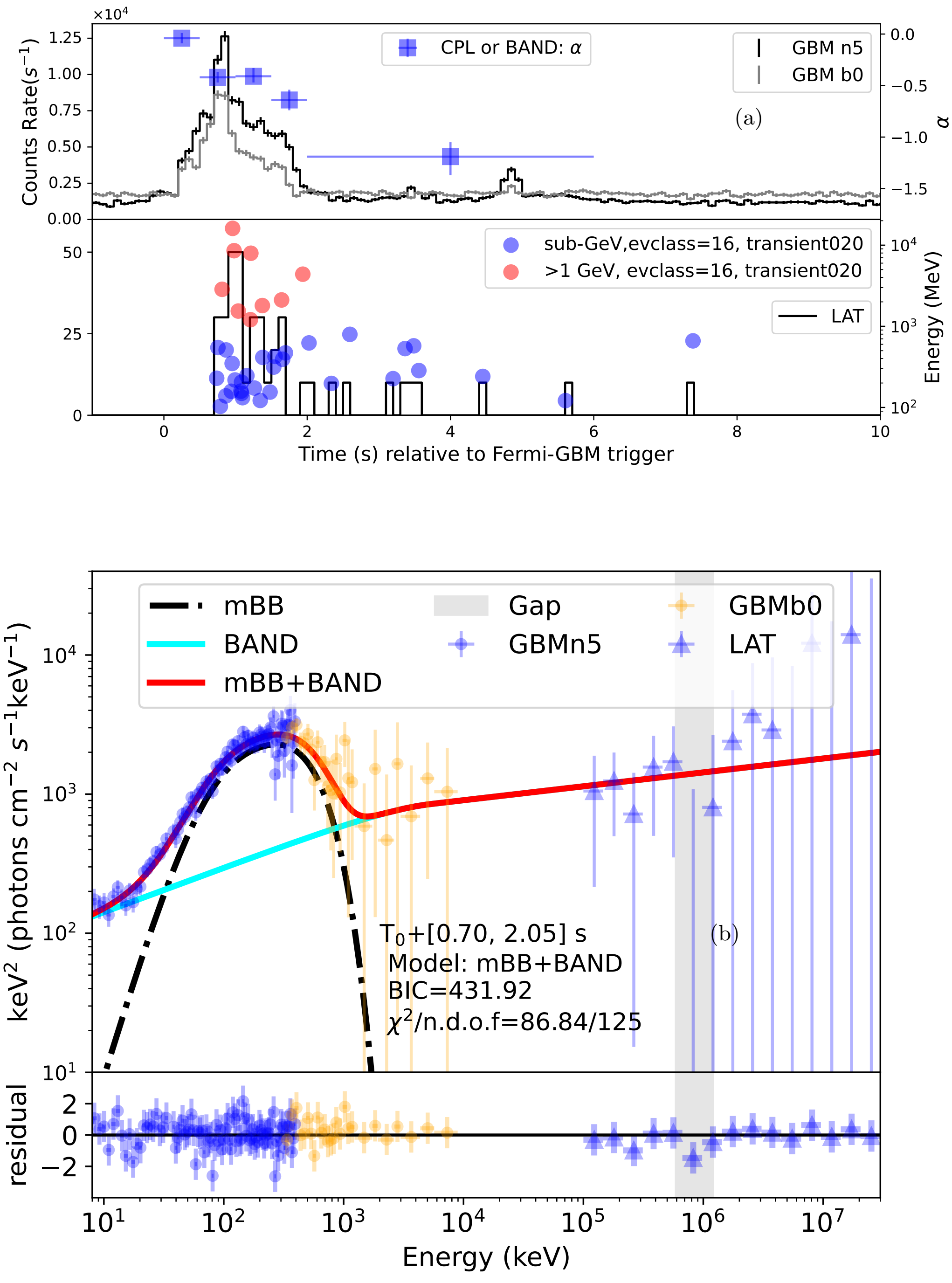

Figure 1. (a) Left Y-axis with the unit of counts rate (s^{-1}): the histograms are the light curves of GBM detectors (NaI 5 and BGO 0, upper plot) and LAT (bottom plot); the high-energy photons are from the LAT data of TRANSIENT020 class (or evclass=16), and used in the analysis to maximize the statistics for spectral analysis of GRB prompt emission. Right Y-axis with the unit of MeV in bottom plot: energies of high-energy photons (denoted by circles). A distinct gap is observed between sub-GeV (blue circles) and GeV (red circles) photons. α values extracted in modeling results with the CPL (or BAND) function are denoted by blue squares with errors. (b) The modeling results with mBB+BAND of $T_0 + [0.7, 2.0]$ s, considering the 0-count bins of the gap.

the keV to GeV energy band; [2] with a lowest BIC and the difference ΔBIC > 10, it is statistically preferred over the other models (see the details in Figure 3 (a)-(c) in Appendix A).[3] As shown in Figure 1 (b), it is evident that the component of a broad energy band has a different origin from that of the peaking component in MeV energy band. It is similar to the case of 090902B (Abdo et al. 2009; Ryde et al. 2010; Pe'Er et al. 2012), therefore we think the peaking component in the MeV energy band is quasi-thermal, and this conclusion is strengthened by the physical interpretation in Section 5.

For the time-integrated (TI) spectrum in $T_0+[0.7, 2.0]$ s, ΔBIC $= 5.6$ is determined for the modelings with mBB+BAND with and without considering the zero-count bins of the gap (as shown in Figure 3 (d) in Appendix A). This means that the preference for the model with the gap is positive.

The estimate of redshift.— The recent multiwavelength study of Pieterse et al. (2026) proposes a collapsar origin with $z = 2.34$; however, Becerra et al. (2023) suggest that the redshift (z) cannot be determined precisely from the detection: some empirical relations, e.g. the Amati relation and the EH-T_{90} diagram, provide a constraint of $z > 0.11$ within the 2σ level (Minaev et al. 2021), thus an estimate of $z < 0.4$ is suggested under the assumption of a short GRB, and their analysis proposed that this burst originates from an exotic high-energy merger. In the first 0.5 s the spectrum is well described by the non-dissipative photospheric (NDP) model (e.g., Lundman et al. 2012; Deng & Zhang 2014; Song & Meng 2022; Song et al. 2022). [4] Note that the estimated $z = 0.27^{+0.14}_{-0.12}$ is extracted by NDP modeling and is model-dependent; it is consistent with the merger hypothesis. In the following, the analysis is presented with $z = 0.27$, while the case of $z = 2.34$ is discussed in parallel and leads to similar conclusions.

The estimate of Γ.— The NDP modeling provides an estimate of $\Gamma \sim 220$ with an uncertainty of about 100 from the MeV emission of the first 0.5 s. From $T_0 + [0.7, 2.0]$ s, Γ could be estimated using the 'top-down' approach (Gao & Zhang 2015; Song et al. 2024).[5] Around the peak (from $T_0 + 0.7$ to 0.9 s), the estimate of Γ ranges between 330 and 270, and drops to about 190 after 1 s. The emission of the most energetic photon (16.1 GeV) around the peak of luminosity requires $\Gamma \gtrsim 200$ (at a radius of $\sim 10^{13}$ cm, determined in Section 4.2), with the optical depth for pair production less than unity. Thus, $\Gamma \approx 200 - 300$ is a reasonable value for GRB 210704A.

For $z = 2.34$, Γ is about 3.2 times that of $z = 0.27$, since $\Gamma \propto \left((1+z)^2 d_L\right)^{1/4}$ (e.g., Pe'er et al. 2007). Moreover, in the following analysis, we find that the conclusions from these two possible redshifts are similar, since the comoving-frame quantities depend on $\Gamma/(1+z)$, which changes only by a factor of approximately 1.2 from $z = 0.27$ to $z = 2.34$.

3. ANALYSIS ON THE ANOMALY IN PROMPT HIGH-ENERGY EMISSION

The probability study of the gap.— High-energy emission above 100 MeV could be described by a PL function. However, it is necessary to investigate whether the gaps between them could be interpreted as statistical fluctuations. The energy spectrum could be described by a continuous PL function with index p,

$$N_{\rm PL}(E) = AE^{-p}. \tag{1}$$

A hypothesis test for the continuous PL modeling of the high-energy emission is performed, and the *null hypothesis* is set to be that the statistical fluctuation could account for the gap of $[0.62, 1.21]$ GeV; the test statistic is the maximum gap below an upper limit of $E_{\rm up} = 1.21$ GeV. A Monte-Carlo (MC) simulation is performed and 1.4×10^5 samples of the same total number of photons are generated based on the TI spectrum of real data in

[2] The parameters are listed as below for mBB+BAND, considering the 0-counts in the gap: $m = -0.37^{+0.55}_{-0.60}$, $kT_{\rm min} = 18.8^{+5.8}_{-4.3}$ keV, $kT_{\rm max} = 145.1^{+7.6}_{-16.8}$ keV, $F_{\rm mBB} = 7.7^{+0.6}_{-0.3} \times 10^{-6}$ erg cm^{-2} s^{-1}, $\alpha = -1.68^{+0.04}_{-0.03}$, $\beta = -1.92^{+0.13}_{-0.07}$, $\log_{10}(E_{\rm p}/{\rm keV}) = 3.89^{+0.14}_{-0.17}$, $F_{\rm BAND} = 21.7^{+2.4}_{-1.7} \times 10^{-6}$ erg cm^{-2} s^{-1}.

[3] Wei et al. (2016) suggested a Bayesian information criterion (BIC$= -2\ln\mathscr{L} + \ln({\rm N}){\rm n}$, $\mathscr{L}$ denotes the likelihood function for all these parameters based on a Bayesian prior, N is the number of data points, and n is the number of free parameters) as a tool for model selection; a model that has a lower BIC value than the others is preferred. If the change of BIC between these two models, ΔBIC is from 2 to 6, the preference for the model with the lower BIC is positive; if ΔBIC is from 6 to 10, the preference for that is strong; and if ΔBIC is above 10, the preference is very strong.

[4] The parameters extracted are listed as follows: $\log_{10} r_0 = 8.53^{+0.16}_{-0.25}$ cm, $\eta = 217.7^{+108.8}_{-80.3}$, $\theta_c = 0.13^{+0.06}_{-0.07}$ and $\log_{10} L_{\rm w} = 49.77^{+0.51}_{-0.65}$, where r_0 is the initial radius for the outflow, η, θ_c and $L_{\rm w}$ are the dimensionless enthalpy, the half-opening angle for the jet core and the wind luminosity, respectively; the index of the power-law index of the profile of structure jet is $p = 8.5^{+6.1}_{-5.9}$, the red shift is determined to be $z = 0.27^{+0.14}_{-0.12}$.

[5] With the method described in Song et al. (2024) and $Y = 2$ (Y is the ratio between the total outflow energy and the energy emitted; the estimated $\Gamma \propto Y^{1/4}$, thus the impact of the value of Y on the estimate of Γ is minor), all bins are found to be in the regime of $r_{\rm ra} \leq R_{\rm ph} \leq r_{\rm c}$, where $R_{\rm ph}$ denotes the photosphere radius, $r_{\rm ra}$ is the rapid acceleration radius, and $r_{\rm c}$ is the coasting radius. The estimated values of Γ have uncertainties of about 50. The initial radius for jet launching is estimated to be $r_0 \sim 10^8$ cm.

$T_0+[0.70, 2.05]$ s; note that the folded model with the response from the detector is used. The probability density (PD) distribution of the maximum gaps is extracted, as shown in Figure 2 (a). The probability of a gap width of ≥ 590 MeV for photons below 1.21 GeV is determined to be $< 10^{-3}$, thus the *null hypothesis* that statistical fluctuation could explain the gap of $[0.62, 1.21]$ GeV is rejected at a significance level of $\alpha = 0.001$ (the corresponding p-value is $< 10^{-3}$).

In fact, the range of the gap varies over time and in the time-integrated spectrum it is the intersection of all, which is the smallest, thus we should consider the smoothing effect of the time-integrated spectrum on the gap; the p-value for the gap is overestimated in the case of time-integrated analysis. Similar procedures are performed in the time-resolved analysis, with the spectral index fixed to that of the time-integrated modeling. As shown in Table 1, the lowest p-value is 6×10^{-4} in $T_0 + [0.9, 1.2]$ s, which is less than that of the time-integrated analysis; the product of the sequence of p-values ($\prod p$-value) is very tiny ($\ll 10^{-6}$) considering the width of the gap that changes over time.

In addition, a full likelihood analysis performed with GTLIKE corroborates this conclusion: comparing the continuous PL model with a model explicitly incorporating a spectral gap of $[0.6, 1.2]$ GeV[6] yields a preference for the model with the gap, with $\Delta\mathrm{BIC} = -2\Delta\log(\mathrm{likelihood}) = 9.2$, which constitutes strong statistical evidence against a simple continuous spectrum.

The possible correlation between $\gamma^{\mathrm{envelope}}_{\mathrm{sub-GeV}}$ and $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$.—The Spearman rank correlation coefficient (ρ) (Spearman 1987)[7] is used to quantify the monotonic relationship of the temporal evolution between the energy of $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$ and that of $\gamma^{\mathrm{envelope}}_{\mathrm{sub-GeV}}$. $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$ is naturally formed by the GeV photons as shown in Figure 2 (b) in red circles connected by a red line. In order to ensure a one-to-one correspondence between $\gamma^{\mathrm{envelope}}_{\mathrm{sub-GeV}}$ and $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$, the maximum sub-GeV energy value in each time bin was extracted as the energy envelope of sub-GeV photons; the data binning in the range of $T_0 + [0.7, 2.0]$ s is performed as follows: 1) the bin edges are determined to be the n-equal diversion points in time of each adjacent GeV point in $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$; 2) to reduce the impact of artificial selection, the bin-edge division ratio (rate) is tested from 0.01 to 0.99 with an increment of 0.01 (for example, a rate equal to 0.2 means left five-division point).

From all binning results with different bin-edge division ratios, four different energy envelopes of sub-GeV photons were extracted, as shown in Figure 2 (b). ρ is determined to be > 0.7 conservatively from the results of all sets of bin edges, which implies a strong correlation between $\gamma^{\mathrm{envelope}}_{\mathrm{sub-GeV}}$ and $\gamma^{\mathrm{envelope}}_{\mathrm{GeV}}$.[8]

4. THE PHYSICAL INTERPRETATION

4.1. *The multi-component emission mechanism, or a two-component jet model ?*

The scenario in which GeV photons are produced by the re-scattering of sub-GeV photons via the inverse Compton (IC) mechanism can hardly account for the energy difference. In the Thomson regime, one has

$$E_{\mathrm{IC,obs}} \simeq \Gamma \gamma_e'^2 E'_{\mathrm{seed}}/(1+z), \tag{2}$$

where the energy ratio of the re-scattered photons to the sub-GeV ones that are only scattered once is $\gamma_e'^2$; $\gamma_e'^2$ is estimated to be about 5 from the observations of (2.86 GeV, 0.55 GeV) and (1.21 GeV, 0.25 GeV). For the production of sub-GeV photons, $E'_{\mathrm{seed}} \sim 500$ keV is too energetic for seed photons in the comoving frame and the Thomson regime. In the Klein-Nishina (KN) regime, re-scattered photons cannot gain more energy via multi-time scattering.

A two-component jet model (a fast spine and slow sheath, e.g. see An 2025) fails to explain the synchronized temporal evolution of the sub-GeV and GeV components. Therefore, we turn to photon-matter interactions.

4.2. *The Resonant Nuclide Absorption Scenario and Other Potential Absorption Mechanisms*

Energy scale.—Applying the Doppler correction $E' \approx E_{\mathrm{obs}}(1+z)/2\Gamma$, the observed gaps translate to a comoving energy range of:

$$E'_{\mathrm{gap}} \approx 2 - 7 \text{ MeV}. \tag{3}$$

[6] For GTLIKE, a "FileFunction" option is allowed for a fixed spectral shape and the sole parameter is a multiplicative normalization. To obtain the best parameter for the PL with a gap of 0.6 to 1.2 GeV, a parameter scan is performed for the index of the power law, and the minimum $-\log(\mathrm{likelihood})$ and the error (the difference between the index of the minimum $-\log(\mathrm{likelihood})$ and that of $-\log(\mathrm{likelihood}) \pm 0.5$) are extracted.

[7] The spearman rank correlation coefficient ρ is defined as $\rho = \frac{\mathrm{cov}(r(x), r(y))}{\sigma_x \sigma_y}$ where $r(a)$ represents the rank of element a. For non-duplicated datasets, ρ can be interpreted as $\rho = 1 - \frac{6\Sigma d_i^2}{n(n^2-1)}$ where $d_i = r(x) - r(y)$.

[8] The grading standards and the corresponding correlation degrees are available online: Spearman's correlation.

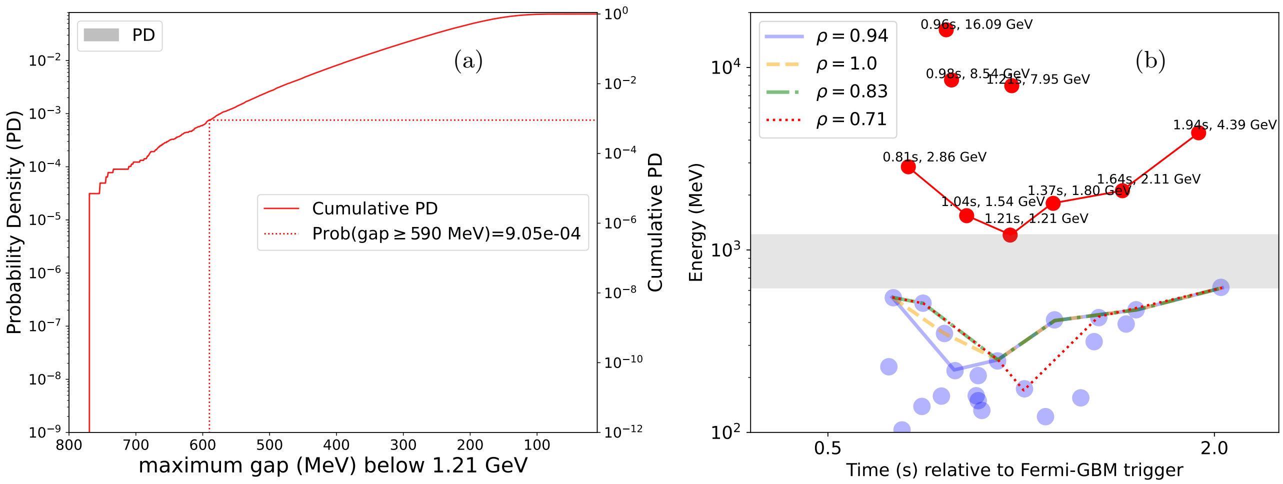

Figure 2. (a) PD (the gray shadow) of maximum gaps in photons below 1.21 GeV and cumulative PD (the solid red line). The p-value of the maximum gap ≥ 590 MeV is labeled by the dotted red line, which is determined by the cumulative PD in red shadow.(b) The Spearman rank correlation coefficients (ρ) of temporal evolution between $\gamma_{\rm GeV}^{\rm envelope}$ and $\gamma_{\rm sub-GeV}^{\rm envelope}$ (denoted by different line styles and colors) extracted with different bin-edge division ratios. The gray shadow between [0.62, 1.21] GeV shows the intersection of all gaps between sub-GeV and GeV photons.

Table 1. Time-resolved gaps and p-values.

Time bins (s)	[0.70, 0.90]	[0.90, 1.20]	[1.20, 1.40]	[1.40, 1.65]	[1.65, 2.05]
Gap range (MeV)	[2860, 550]	[1540, 350]	[1210, 410]	[2110, 430]	[4390, 620]
p-value	0.0037	0.0006	0.0180	0.0067	0.0034

Note that for $z = 2.34$ the correspondingly larger Γ leads to a similar range, since $\Gamma/(1 + z)$ changes only by a factor of about 1.2 (see Section 2).

We propose that the gap arises from the resonant absorption of photons by nuclei entrained in the jet. The comoving energy of ~ 4 MeV aligns remarkably well with the photodissociation threshold of deuterium:

$$\gamma + {}^2\mathrm{H} \rightarrow p + n. \quad (4)$$

The binding energy of deuterium is 2.22 MeV. However, the cross-section for photodissociation peaks slightly above the threshold, typically around $4 - 5$ MeV (International Atomic Energy Agency 2000), which perfectly matches our derived comoving energy $E'_{\rm gap}$.

Crucially, this energy scale distinguishes deuterium from all other potential nuclear absorbers. Heavier nuclides, particularly the Fe-group elements often invoked in supernova models, interact with photons primarily through the Giant Dipole Resonance (GDR). The GDR peak energy for nuclei with mass number A is approximately $E_{\rm GDR} \approx 31.2A^{-1/3}+20.6A^{-1/6}$ MeV (Berman & Fultz 1975), which places the resonance at $15 - 25$ MeV for heavy elements (e.g., ^{56}Fe) (Kawano & et al. 2020). Such an absorption would manifest at observed energies > 3 GeV, far above the gap detected in GRB 210704A. The unique low-energy threshold of deuterium makes it the only viable candidate for an absorption feature at ~ 4 MeV.

Timescales and Nucleosynthesis.—The identification of deuterium is further supported by the dynamical timescales of the GRB jet. With the standard internal shock model for the prompt emission, the emitting radius is (Zhang & Yan 2011)

$$R \simeq 2\Gamma^2 c\delta t/(1+z) \simeq 6\times 10^{12}\Gamma_{2.5}^2\delta t_{-3}/(1+z)\ \mathrm{cm}, \quad (5)$$

where δt is the timescale of the pulse duration; hereafter, the convention $Q_s = Q/10^s$ is adopted in cgs units throughout the text. δt is determined to be about 1 ms by a wavelet decomposition of the light curve of the internal shock emission (photons with energies < 20 keV are required, and the contribution from the quasi-thermal emission is 7%; the light curve has a sampling rate of 4000 Hz; see the details of the method in MacLachlan et al. 2013). The emitting radius is thus of the order of $10^{12} - 10^{13}$ cm, a typical value for short GRBs (Berger 2014). The dynamical time in the comoving frame is then extremely short, only a fraction of a second:

$$t'_{\rm dyn} \approx \frac{R}{\Gamma c} \sim 0.1 R_{12} \Gamma_{2.5}^{-1} \text{ s.} \tag{6}$$

In a neutron-rich outflow ejected from a merger, nucleosynthesis begins with the rapid capture of neutrons by protons ($p + n \rightarrow {}^2$H). To form heavier r-process elements (up to lanthanides), the chain must proceed through seed nuclei creation. However, the formation of seeds heavier than deuterium is rate-limited by weak interactions (β-decays), which typically require timescales of seconds to minutes to proceed significantly; a fraction of a second is far too short for heavier elements to form.

Given the relativistic expansion, the jet material is "frozen out" before significant quantities of heavy elements can form via β-decay. Consequently, the jet composition at the photospheric radius is expected to be dominated by nucleons and light isotopes, specifically deuterium, rather than heavy metals. This kinetic constraint robustly excludes Fe-group elements and reinforces the deuterium interpretation.

Other candidate absorption mechanisms.—As another candidate absorption mechanism, photon-photon pair production ($\gamma + \gamma \rightarrow e^+ + e^-$) could in principle attenuate the high-energy spectrum; however, its optical depth increases monotonically with the photon energy above the threshold, so it would produce a high-energy cutoff rather than a discrete gap followed by a recovery in flux above 1.21 GeV. Moreover, to generate an observed absorption with a center energy of ~ 1 GeV, pair production requires a much larger Lorentz factor, e.g. $\Gamma \gtrsim 1000$, than the estimate. Therefore, it is unlikely unless there exists sufficient evidence for an extremely large Lorentz factor.

5. DISCUSSION AND CONCLUSION

Discussion.— The comoving density of the jet, mainly contributed by protons, is expressed in terms of the isotropic kinetic energy of the jet $E_{\rm K}$ at the emission radius R as [9]

$$\begin{aligned} n'_p &= \frac{E_{\rm K}}{4\pi R^3 m_p c^2} \\ &= 1.7 \times 10^{17} E_{\rm K,53} R_{12.5}^{-3} \text{ cm}^{-3}, \end{aligned} \tag{7}$$

[9] In the comoving frame, the shell volume is $V' = 4\pi R^2 (R/\Gamma)$; one has $n'_p V' m_p c^2 = M_{\rm iso} c^2$ and $E_{\rm K} = \Gamma M_{\rm iso} c^2$.

where $E_{\rm K,53} = E_{\rm K}/10^{53}$ erg and the unit of R is cm. The comoving density of deuterium n'_D is estimated with the number fraction of deuterium (Y_D) to be:

$$n'_D = 0.1 n'_p Y_{D,-1}. \tag{8}$$

The optical depth for the photodissociation of deuterium is

$$\tau_{\gamma D} \sim \frac{\sigma_{\gamma D} n'_D R}{\Gamma} = 1.3 E_{\rm K,53} R_{12.5}^{-2} Y_{D,-1} \Gamma_2^{-1}, \tag{9}$$

where $\sigma_{\gamma D} \simeq 2.5 \times 10^{-27}$ cm^2 at a photon energy $\varepsilon'_\gamma \simeq 4.48$ MeV in the deuteron rest frame (International Atomic Energy Agency 2000).

Internal shocks are natural sites for the dissipation of the kinetic energy of a baryonic fireball (Shemi & Piran 1990; Rees & Meszaros 1992; Meszaros & Rees 1993; Rees & Meszaros 1994). In the internal shock model of gamma-ray bursts, electrons accelerated at mildly relativistic shocks are commonly assumed to follow a non-thermal distribution (Piran 2004),

$$N(\gamma_e) \propto \gamma_e^{-p}, \qquad \gamma_e \geq \gamma_m, \tag{10}$$

where γ_m is the minimal Lorentz factor of the electrons accelerated in the internal shocks. For typical internal shock parameters adopted in the literature, $\gamma_m \sim 10^{3.5}$–10^6 (Beniamini & Piran 2013; Oganesyan et al. 2019). The scattering regime between photons and relativistic electrons is determined by

$$x \equiv \frac{\gamma_e \varepsilon'_\gamma}{m_e c^2}, \tag{11}$$

where ε'_γ is the photon energy in the comoving frame. One has $x \gtrsim 2.5 \times 10^4$ for $\varepsilon'_\gamma = 4$ MeV and $\gamma_m \gtrsim 10^{3.5}$. In this Klein-Nishina limit, the total scattering cross section is suppressed to (Rybicki & Lightman 1986)

$$\sigma_{\rm KN} \simeq \frac{3}{8} \sigma_T \frac{\ln(2x) + 1/2}{x} \lesssim 1.1 \times 10^{-28} \text{cm}^2, \tag{12}$$

thus, the optical depth for Compton scattering in the Klein-Nishina regime is

$$\tau_{\rm KN} \sim \frac{\sigma_{\rm KN} n'_e R}{\Gamma} \lesssim 0.6 E_{\rm K,53} R_{12.5}^{-2} \Gamma_2^{-1}, \tag{13}$$

assuming local electric neutrality in the jet, $n'_e \simeq n'_p$. The break energy (ε_b) in the spectrum of the non-thermal component described by a BAND function (e.g. the synchrotron emission by the relativistic electrons in magnetic field) may correspond to $\tau_{\rm KN}(\varepsilon'_b) \sim 1$ at $\gamma_e = \gamma_m$. From the results of the modeling of the non-thermal component, the break energy is $\varepsilon_b \sim 10^4$ keV, which corresponds to $\gamma_m \sim 10^{5.4}$ in this case;

$\sigma_{\rm KN} \sim 10^{-30}$ cm^2 is much lower for the photons of $\varepsilon'_\gamma = 4$ MeV, due to a deeper Klein-Nishina regime.

Let us demonstrate the plausibility of the observation of photodissociation of deuterium in the jet of GRB 210704A as follows: the first key point is whether the baryon loading in the relativistic outflow could produce enough deuterium; second, GeV photons are emitted while those in the resonant region are absorbed by deuterium, which requires that $\tau_{\gamma D}$ and $\tau_{\rm KN}$ (for GeV photons) satisfy

$$\tau_{\rm KN} < 1 \lesssim \tau_{\gamma D}. \quad (14)$$

The freezeout Y_D can be greater than a few percent or even as large as 10% under the right conditions below the radius of GeV emissions (see the simulation in Pruet et al. 2002); the production of deuterium could be further enhanced through the spallation of α-particles (Beloborodov 2003), which is caused by neutron-ion decoupling as well as internal shocks. Therefore $\tau_{\gamma D}$ could be larger than $\tau_{\rm KN}$ in this case. The total energy of the outflow could be estimated from the radiated energy $E_{\gamma,\rm iso}$ and the radiative efficiency (ϵ_γ) as

$$E_{\rm tot,iso} = E_{\gamma,\rm iso}/\epsilon_\gamma; \quad (15)$$

for baryonic fireball, $E_{\rm K} \lesssim (1-\epsilon_\gamma)E_{\rm tot,iso}$. It is reasonable that $\epsilon_\gamma \lesssim 10\%$, because the radiation efficiency of internal shocks is typically low (Zhang & Yan 2011; Fan & Piran 2008). $E_{\gamma,\rm iso} \simeq 10^{52}$ erg for GRB 210704A, thus we assume that $E_{\rm K}$ is comparable to 10^{53} erg, or even several times larger. The internal shock radius (Eq. 5) has an order of magnitude of $10^{12} - 10^{13}$ cm. Thus, given $Y_D \simeq 0.1$, $E_{\rm K} \sim 10^{53}$ erg and $R \sim 10^{12.5}$ cm, Eq. (14) is satisfied. The baryon loading in the jet is calculated to be $f_b M_{\rm iso} \sim 10^{-6} M_\odot$ by $E_{\rm K} = \Gamma M_{\rm iso} c^2$ (the typical beaming factor for short GRBs is $f_b \sim 6 \times 10^{-3}$ (Berger 2014)), which is a reasonable value for merger events (Metzger 2019; Song 2025).

Conclusion.— We identified a statistically significant spectral gap in the prompt emission of GRB 210704A. Interpreting this for the first time as resonant absorption by deuterium nuclei entrained in the relativistic jet, we find a consistent physical picture that links the burst to a neutron-rich outflow. This represents the first direct spectroscopic detection of r-process nucleosynthesis ingredients in the prompt gamma-ray phase, opening a new window for studying nuclear astrophysics in extreme environments.

Previous studies of GRB 210704A noted its peculiar properties, with its duration lying in the intermediate range; both a compact-object merger origin (Malacaria et al. 2021; Becerra et al. 2023) and a collapsar origin (Pieterse et al. 2026) have been proposed. Unlike kilonovae, which probe the sub-relativistic ejecta days after the event, this spectral feature probes the relativistic jet itself seconds after launch, revealing the initial chemical composition of the r-process flow. It should be noted that such observations are expected to be rare: GRB 210704A is a 'short' long burst that favors the abundance of deuterium, and the high baryon loading of the outflow together with the rapid variability of the central engine creates the conditions for the occurrence of intense internal shocks.

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APPENDIX

A. SPECTRAL MODELINGS WITH DIFFERENT MODELS

The modeling results with the other models, including BAND, BB+BAND and mBB+PL, are shown in Figure 3 (a)-(c). The modeling result with mBB+BAND without the zero-count bins in the gap is shown in Figure 3 (d).

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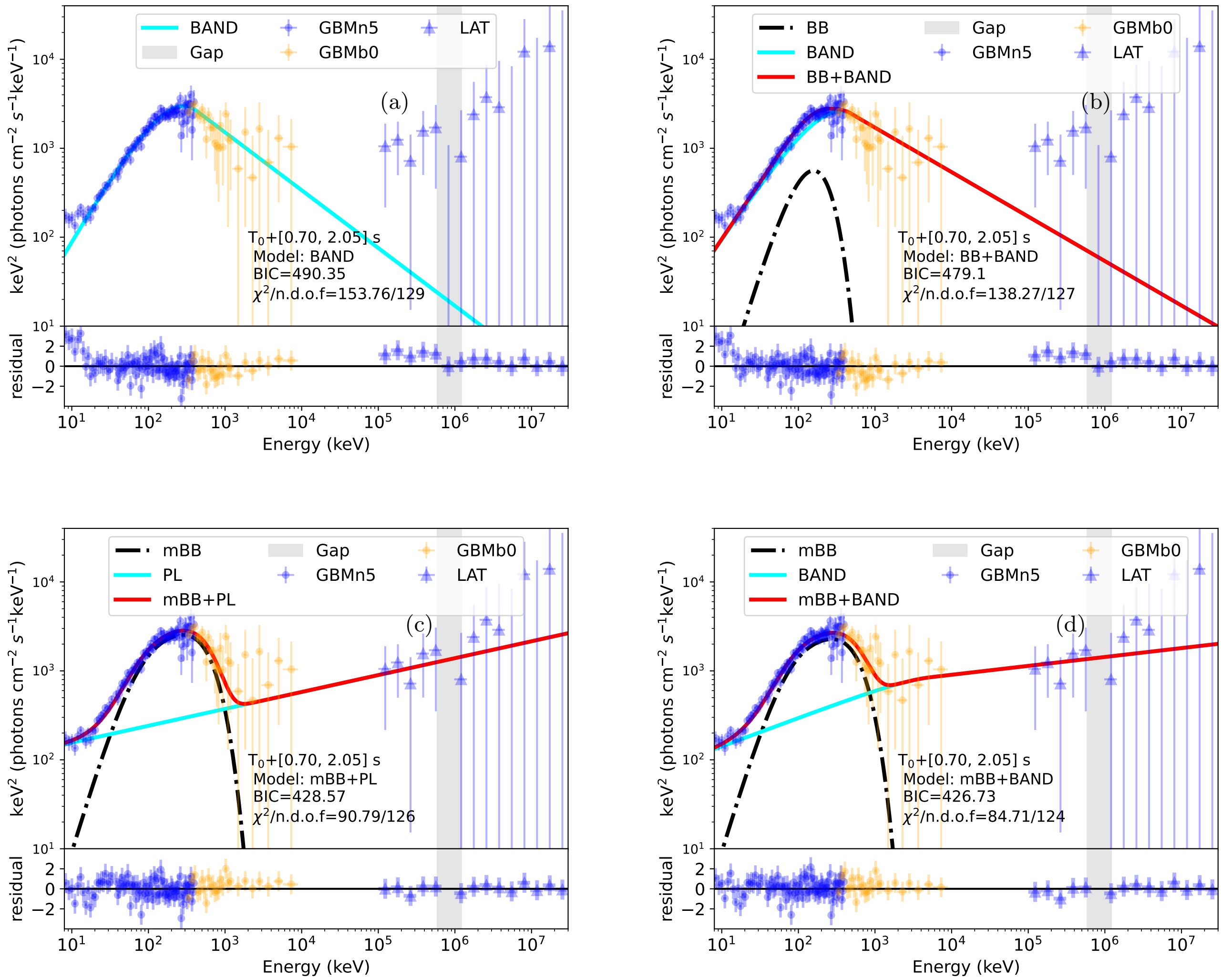

Figure 3. Modeling results with BAND, BB+BAND, mBB+PL and mBB+BAND (modeling without zero-count bins in the gap).

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